1. Introduction

Text-to-image diffusion models can generate highly realistic images from natural language prompts, but they often struggle to preserve the identity of a subject across different generated scenes. While pretrained models are trained on billions of images, without alteration they don't understand the specific, nuanced appearance of a user's personalized object. Our project reproduces and evaluates the results of *DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation* (Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, Kfir Aberman)*,* a method that personalizes a pretrained diffusion model fine tuning it on a small set of subject images. The method detailed in the paper updates weights of the diffusion model to associate a unique identifier with the given subject, using subject images to learn information about the subject and class images to keep the model's understanding of the general class intact. As a result, the paper contributes a personalized, subject-driven fine-tuning technique for text-to-image diffusion models that allows a model to learn a specific, novel subject from only a few (3–5) images, and subsequently place that subject in diverse, new contexts. This work is motivated by the growing use of personalized generative AI in creative design, content generation, and human-centered applications.

2. Chosen Result

Given this set of subject images, our goal was to reproduce the results of generating novel images of this chosen subject in new contexts while maintaining its identity–for example, through recontextualization,accessorization, art renditions, and property modification. Reproducing subject and prompt fidelity is important because the main overarching goal of the original paper was to be able to maintain both in image generation, an improvement from base diffusion models that have good prompt fidelity but lack subject fidelity. We use the same metrics (DINO, CLIP-I, and CLIP-T) to evaluate the quality of the generated images in terms of both subject fidelity and prompt fidelity, and aim to achieve comparable metrics. By tracking both subject fidelity metrics (DINO and CLIP-I) and prompt fidelity (CLIP-T), we aren't just guessing if the generated image follows the prompt well or resembles the original subject well–we can quantitatively evaluate and compare performance.
From the original paper:

Method	DINO ↑	CLIP-I ↑	CLIP-T ↑
Real Images	0.774	0.885	N/A
DreamBooth (Imagen)	**0.696**	**0.812**	**0.306**
DreamBooth (Stable Diffusion)	0.668	0.803	0.305
Textual Inversion (Stable Diffusion)	0.569	0.780	0.255

Table 1. Subject fidelity (DINO, CLIP-I) and prompt fidelity (CLIP-T, CLIP-T-L) quantitative metric comparison.

3. Methodology

We reimplemented DreamBooth by using Stable Diffusion v1.5 as the base generative model and fine-tuning it, where the original paper used Imagen. This has three main components: VAE, CLIP Text Encoder and UNet. To fit training within Colab GPU memory, we froze the VAE, text encoder, and spatial layers of the UNet. Fine-tuning was restricted primarily to the cross-attention layers in UNet. This partially frozen approach significantly reduced the number of trainable parameters while still allowing the model to learn the new subject identity.

Our training dataset consisted of two components: 5 images of Adelynn's water bottle captured from different viewpoints and lighting conditions as subject images and 100 generic water bottle images generated using the base Stable Diffusion v1.5 model as class prior images. The subject images were paired with prompts containing a unique identifier token, [V].

Training followed the standard diffusion fine-tuning process. The objective is total loss = subject loss + lambda * prior preservation loss, where subject loss encourages accurate generation of the target subject associated with the token [V], and prior preservation loss maintains the model's ability to generate realistic images. We varied prior preservation weights lambda = 0, 0.25, 0.5, 0.75, and 1.0, and trained the model for multiple iterations while monitoring generated samples to evaluate identity preservation, visual quality, and prompt consistency.

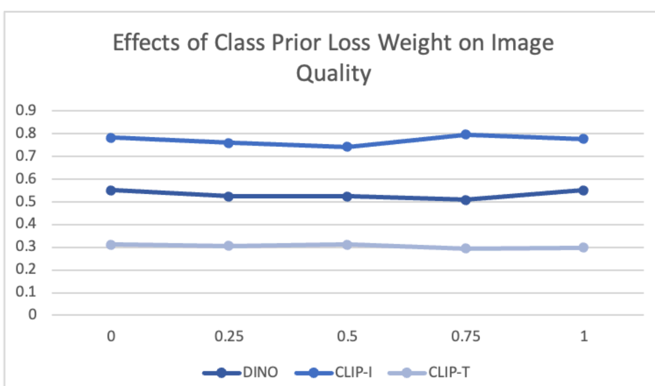
For evaluation, we follow the DreamBooth evaluation idea: generate four images per prompt, then compute DINO and CLIP-I for subject fidelity and CLIP-T for prompt fidelity.

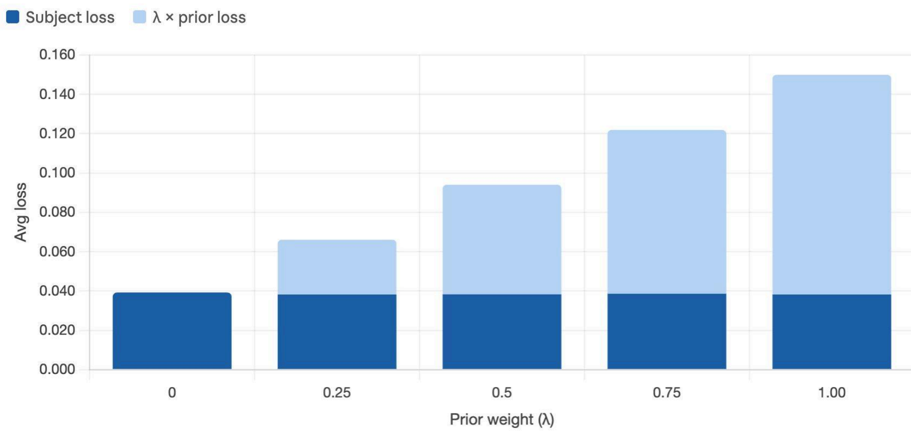
4. Results & Analysis

Compared to the results in the paper, our DINO score was lower. This indicates that our model struggled to preserve certain fine-grained details. However, our CLIP-I score slightly exceeded the reference values, suggesting that generated images remained semantically similar to the subject in CLIP embedding space. Our CLIP-T score was also slightly higher than the paper's values, indicating that the model followed text prompts effectively.

	DINO	CLIP-I	CLIP-T
DreamBooth (Imagen)	0.696	0.812	0.306
DreamBooth (SD)	0.668	0.803	0.305
Reimplementation (SD)	0.554	0.826	0.313

To further analyze our implementation, we evaluated the effect of varying the prior preservation loss weight from 0 to 1. Across all settings, DINO, CLIP-I, and CLIP-T scores remained relatively stable. Loss analysis showed that increasing the prior preservation weight mainly scaled the total loss, while the subject loss remained relatively unchanged throughout training.





This result differs somewhat from full DreamBooth training, where prior preservation is critical for preventing overfitting and class drift. During implementation, the primary challenge was GPU memory limitations. To address this, we froze most of the Stable Diffusion backbone. The model already retained strong general knowledge of subject class. Thus, subject learning was driven primarily by the instance images, identifier token [V], and the fine-tuned cross-attention layers rather than the prior preservation term.

Overall, these findings suggest that our partially frozen DreamBooth pipeline can achieve reasonable subject-conditioned generation with little or even no prior preservation loss. While it sacrifices some fine-grained identity fidelity compared to full DreamBooth, it significantly reduces computational cost and memory requirements.

5. Reflections

One of our main lessons was that reproducing generative models often requires balancing fidelity, training stability, and hardware limitations. We also learned the impact of implementation details in research papers and the importance of iterative experiments. In our case, how many prior images to use and how to choose the prior loss weight affected the training behavior and output quality. A major takeaway from our project is that freezing most of the Stable Diffusion backbone changes the role of prior preservation. In full DreamBooth, prior preservation is necessary to prevent the model from overfitting. However, because our implementation kept most pretrained weights fixed, the model already retained much of its general understanding.

For future work, we would like to explore parameter-efficient fine-tuning methods such as LoRA, which could further reduce memory usage while improving subject fidelity. We would also test larger and more diverse subject datasets, experiment with different base models, and compare different personalization strategies beyond DreamBooth.

6. References

[1] Ruiz, Nataniel et al. “DreamBooth: Fine Tuning Text-to-Image Diffusion Models for Subject-Driven Generation.” 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) (2022): 22500-22510.